

Task 5: Classifier-Free Guidance (Encoder-Decoder)

1. Objective

Classifier-Free Guidance (CFG) aims to steer the generation process toward higher quality outputs by using a secondary classifier's gradients. This task evaluates whether a 139K parameter MLP can improve the semantic fidelity of the Encoder-Decoder variant at T=4.

 **Observed Status:** Implementation complete. However, the current Sanskrit training corpus (40-60 samples for the classifier) remains insufficient to provide stable gradients, leading to $\lambda=0.0$ (guidance disabled) being the optimal production scale.

2. Technical Architecture

A 3-layer MLP was trained to predict BERT-F1 semantic scores from the latent states of the Encoder-Decoder bottleneck.

 **Implementation:** [analysis/quality_classifier.py](#)

```
# CFG Logit Modification Logic:
if guidance_scale > 0.0:
    logit_grad = classifier.get_gradient(hidden)
    logits = logits + guidance_scale * logit_grad
```

3. Results Discussion

For the Encoder-Decoder model, the latent representations showed higher variance than the cross-attention variant. This introduced additional noise during the CFG gradient injection, making the model more sensitive to the guidance scale λ . At $\lambda > 1.5$, the model quickly collapsed into repetitive punctuation or out-of-vocab tokens.

Classifier Parameters

139.14K

Optimal Guidance Scale

(λ)

0.00

4. Future Work

To enable effective CFG for this architecture, future work should focus on scaling the "Quality-Labeled" dataset from 40 samples to >2,000 samples to allow the Quality Classifier to learn more robust semantic boundaries.